Reproducibility map for "What Prices Reveal About Hidden Trader Populations, and What Identifying Them Costs"

Source of every number

The replication package contains numbered Colab notebooks and stand-alone scripts; its README lists them in run order. "Part N" below and in table notes refers to notebook N (Part 1: theorem checks and spot tests; Part 2: BTCUSDT futures depth; Parts 4–13: the cross-section, replications, fresh-day tests, second venues, the CFTC test, the certified guard and post-hoc checks). Result identifiers such as `X6` or `U6` name the JSON file that the notebook writes. "Kind" distinguishes closed-form or simulation results (recomputed), results from public market data (real data), and audits.

Result	Source		Kind
Complete monotonicity, 892 economies	`L26_complete_monotonicity`		recomputed
$d_2 = -\kappa^2 q \delta^2$	`P1.7-T1_third_tap`		recomputed
Ledger identity; T_h positive definite	`R2-T1`, `R2-T3`		recomputed
Figure S6	`make_figs2.py`		recomputed
$\delta_{\min}(3)$, Table S12 recomputed columns, 61.14, 29.94	`recompute_theory.py` `R_P1_theory_constants.json`	→	recomputed
Pulse benchmarks (271, 58, 12, 11 pulses), Figures S7–S1	`make_figs3.py` `R_P1_pulse_benchmarks.json`	→	recomputed
Table S32, Table S33, worked example, passive simulation	`recompute_extra.py` `R_P1_extra.json`	→	recomputed
Daily noise (Table S15)	`P1.3-T1_sigma_obs`		real data
Table S30 recomputed columns	same		recomputed
K-cohort slopes (Table S34)	`T1_K_cohort_rate`		recomputed
ESPRIT slope −6.15	`P1.2-T1_esprit_rate`		recomputed
Minimum phase, 1,733 economies; rank 6 of 6	`T2a`, `T2b`		recomputed
Passive slopes −4.21, −5.93	`T3_passive_rate`		recomputed
Tagged-flow slope 0.011	`T4_disclosure`		recomputed
Spectral blindness gains	`T5_spectral_blindness`		recomputed
Noise and capital grid (Table S14)	`P1.3-T1_sigma_obs`		real data

Result	Source		Kind
Loop gain and placebo (Table S20)	`P1.6-T1_ema_placebo`		real data
Estimator table (Table S21)	`P1.5-T1`, `T2`, `T3`		recomputed
Data integrity (111 files)	`R2-T9_data_integrity`		audit
Tables S11, S12 (certified, feasible, adaptive columns)	`recompute_p1_capital.py` `R_P1_capital_ledger.json`	→	recomputed
Table 1; resolution floors; Table S2	`recompute_p1_protocols.py` `R_P1_protocols.json`	→	recomputed
SPRT capital figures	`recompute_p1_sprt.py` `R_P1_sprt.json`	→	recomputed
Table S9 (N^*, σ_{crit})	`recompute_crb.py` `R_P1_crb.json`	→	recomputed
Table S9 (ML columns), Table S30 (estimators)	`recompute_p1_estimators.py`		recomputed
Symmetric multi-asset network (Supplement S12)	`recompute_oct20.py`		recomputed
Proof audit: K_* bound and d_2 identity (Theorem 4.1), moment gaps (Theorem S25.1), spectral-blindness example (Theorem S12.1)	`recompute_audit.py` `R_audit_oct2026.json`	→	recomputed
Theorem 5.2, Table S35	`kcohort_upper.py` `R_P1_kcohort_upper.json`	→	recomputed
Lem. S17.1, Thm S17.2, Tab. S24	`general_minimax.py`		recomputed
Theorem 6.2 numbers	`disclosure_value.py` `R_P1_disclosure_value.json`	→	recomputed
Tables S18, S19, S22	Part 8: `V2_spot_panel`, `V3_fee_events`		real data
Table S27, top panel	Part 4: `X9_disclosure_metrics`		real data
Table S27, bottom panel; class-position forecasts	Part 6: `Z6_C_identification`, `Z6_secondary`; `P6_fresh_days.json` (65acda36…)		real data
Table S25	Part 11: `C3_cot_data`, `C4_cot_disclosure`; `P11_cot.json` (6f1d6f35…)		real data

Result	Source	Kind
Basis-trade signature; real-data Cochrane–Orcutt	Part 13: `R6`, `R7`	real data (post hoc)
Feedback across 30 perpetuals	Part 4: `X3_T6_cross_section`	real data
Theorem S9.4 full bound (Table S13), direct minimization	`hardening/rh_bound.py`, `rh_check.py` → `R_P1_rh_explicit.json`	recomputed
Minimal dimension at $a = 0$; Toeplitz sections; passive slopes for $K \leq 4$; IV determinant	`hardening/p1_checks.py` → `R_P1_hardening_checks.json`	recomputed
Stylized nonlinear and trained online rules (Table S16, Figure S4)	`recompute_p1_learners.py` → `R_P1_learners.json` (Panel A); `validate_trained_learners_hardened.py` → `R_P1_trained_learners_hardened.json` (Panels B–C); specification `P1_trained_learners.json` (3e8bf30c…)	recomputed
Disclosure in dollars (Table S10)	`recompute_p1_cat_costs.py` → `R_P1_cat_disclosure_dollars.json`	recomputed
Out-of-sample post-shock return test (Supplement S19)	`recompute_prereg_oct21_22nd.py` → `R_P1_disclosure_oos.json`; `P1_disclosure_oos.json` (f0182ce5…)	real data
Out-of-sample cohort-state trajectory test (Table S28)	`recompute_prereg_oct21_followup.py` → `R_P1_disclosure_state_oos.json`; `P1_disclosure_state_oos.json` (5136cc6b…)	real data
Proposition 6.4, Table S36	`theory/targeted_intervention.py` → `R_P1_targeted_intervention.json`	recomputed
Table S30, column N^*_{CRB}; Table S9, spacing-0.10 three-cohort row	`breakthrough_experiments/results_exp17_minimax_lecam_bou` `results_exp15_cohort_information.json` (canonical run)	recomputed

Result	Source	Kind
Table S13, best feasible probes and $L = 320, 480$	`R_P1_capital_ledger.json`, `R_P1_cquadL.json` (reproduced by `hardening/rh_quad.py`)	recomputed
Table S11, peak margins of the eigenvector probes	`breakthrough_experiments/results_exp14_capital_frontier.json`	recomputed (earlier run)
Table S24	`theory/R_P1_general_minimax.json`	recomputed
Table S30 and Table S9, estimator columns	`R_P1_estimators.json`, `R_P1_estimators_crb.json`	recomputed
Theorem 3.1, numerical check of the equivalence	`R2-T5_merged_invariance`	recomputed
Supplement S19 and Table S29: horizon-matched test on 30 new markets (29 of 29, median 7.3, smallest 2.6) and jackknife-corrected persistence (1.0008)	`thesis/evidence/C2_results.json` (specification `thesis/checks/prereg/C2_margin_vs_identify.json`)	real data
Supplement S19: exploratory 48- and 96-lag refit (median 7.5)	`thesis/evidence/pilot3_horizon_matched_results.json`	real data (exploratory)
Supplement S19 and Table S29: 105-day follow-up (median ratio 74.9; 9 identified; 10 of 30, 23 required)	`thesis/evidence/E1v2_results.json`	real data
Supplement S19 and Table S29: 58-day follow-up (22 of 29; 20 of 29, 22 required)	`thesis/evidence/D1_results.json`	real data
Supplement S16, Tables S23, S3 and S29: real-noise injection (slopes -6.01, -9.81; tagged 256–724)	`p1_injection/injection_test.py`, `p1_injection/results/I1_injection_test.json` (specification `p1_injection/prereg/P1_injection.json`, 596a700d…)	semi-synthetic
Supplement S3.7 and Table S4: numerical stress tests	`p1_stress/stress_theorems.py`, `p1_stress/results/S1_stress_tests.json`	numerical check

Result	Source	Kind
Supplement S3.8 and Table S5: probing a running market without resets	`p1_running/running_market.py`, `p1_running/results/R1_running_market.json`	numerical check
Theorem 3.1(i), Theorem 5.1 and Proposition 6.1 (machine-checked parts)	`research_extension/current_manuscript_lean_20261008/CurrentProofs/Paper1MergedModel.lean`, `Paper1Identification.lean`, `Paper1MomentGap.lean`, `Paper1PoleCount.lean`, `Paper1FiniteIdentification.lean`, `Paper1Tagged.lean`, `Paper1Testing.lean`, `Paper1LowerBound.lean`, `Paper1MatchedPair.lean`	Lean 4 proof
Theorem 5.2 (machine-checked, every K)	`research_extension/current_manuscript_lean_20261008/CurrentProofs/Paper1PronyUpper.lean`, `Paper1PronyMoment.lean`, `Paper1PronyDeconv.lean`, `Paper1UniformInverse.lean`	Lean 4 proof
Theorems 3.1(ii), S12.1, 5.4 (zeros), 6.2(ii) (tagged upper bound) and Proposition 6.1 (instrument determinant), machine-checked parts	`research_extension/current_manuscript_lean_20261008/CurrentProofs/Paper1MergedSeparation.lean`, `Paper1Blind.lean`, `Paper1MinPhase.lean`, `Paper1TaggedRates.lean`, `Paper1TaggedIV.lean`	Lean 4 proof

Result	Source	Kind
Theorem 5.5(i) and (iii), sample bounds for passive data from rest after any warm-up, any moment-matched pair (machine-checked)	`research_extension/current_manuscript_lean_20261008/CurrentProofs/Paper1PassiveCost.lean`, `Paper1PassiveKCohort.lean`, `Paper1PassiveMatched.lean`, `Paper1PassiveRecord.lean`, `Paper1GaussianTriangular.lean`	Lean 4 proof
Theorem 5.4, model links: poles of H are the eigenvalues of the market transition, and the recursion's impulse response has transfer function H (machine-checked)	`research_extension/current_manuscript_lean_20261008/CurrentProofs/Paper1ModelLink.lean`	Lean 4 proof
Theorem 5.1: the three-point Gauss and Gauss–Radau rules, their matched moments and fifth-moment gap (machine-checked)	`research_extension/current_manuscript_lean_20261008/CurrentProofs/Paper1GaussRadau3.lean`	Lean 4 proof
Theorem S17.2, both parts (machine-checked)	`research_extension/current_manuscript_lean_20261008/CurrentProofs/Paper1GeneralMemoryLower.lean`, `Paper1Jackson.lean`, `Paper1JacksonPoly.lean`, `Paper1GeneralMemoryUpper.lean`, `Paper1GeneralMemoryRate.lean`	Lean 4 proof
Proposition 6.4, supervisory separation (machine-checked for two groups; part (i) for every K)	`research_extension/current_manuscript_lean_20261008/CurrentProofs/Paper1SupervisorySeparation.lean`	Lean 4 proof

Result	Source	Kind
Theorem 4.1(i) and Lemma S24.4, quantity and capital lower bounds (machine-checked)	`research_extension/current_manuscript_lean_20261008/CurrentProofs/Paper1QuantityCapital.lean`	Lean 4 proof
Pinsker's inequality for events; Theorems 5.1, S25.1, 4.1(i) and Lemma S24.4 at every error level (machine-checked)	`research_extension/current_manuscript_lean_20261008/CurrentProofs/Paper1Pinsker.lean`, `Paper1GeneralLevel.lean`	Lean 4 proof
Theorem S3.1, finite-window abstention (machine-checked)	`research_extension/current_manuscript_lean_20261008/CurrentProofs/Paper1Abstain.lean`	Lean 4 proof
Corollary 5.3(i)–(ii), identification with unknown (a, κ) (machine-checked)	`research_extension/current_manuscript_lean_20261008/CurrentProofs/Paper1UnknownPrimitives.lean`	Lean 4 proof
Proposition S7.1(i), Cramér–Rao bound on the raw records of the actual market (machine-checked)	`research_extension/current_manuscript_lean_20261008/CurrentProofs/Paper1GaussianHCR.lean`, `Paper1GaussianCR.lean`, `Paper1CRActual.lean`, `Paper1CRRaw.lean`	Lean 4 proof
Proposition S7.1(iii), two-sided $(\Delta\eta)^{-6}$ bound on both weight entries of the inverse Fisher matrix, fixed $T \geq 4$ (machine-checked)	`research_extension/current_manuscript_lean_20261008/CurrentProofs/Paper1FisherBounds.lean`, `Paper1CRTaps.lean`, `Paper1CRS71iii.lean`, `Paper1CRS71iiiEta.lean`, `Paper1CRS71iiiW2.lean`	Lean 4 proof

Result	Source	Kind
Theorem 3.4 (equilibrium invariance) under the merged measure (machine-checked pathwise)	`research_extension/current_manuscript_lean_20261008/CurrentProofs/Paper1EquilibriumInvariance.lean`, `ceiling_suite/papers/paper1/equilibrium/e1_demo.py`	Lean 4 proof; numerical illustration
Theorem 5.4(ii) for every K: the spectrum determines $\sigma_\xi^2\kappa^2$, H/h_0 and, with κ known, the merged measure (machine-checked)	`research_extension/current_manuscript_lean_20261008/CurrentProofs/Paper1SpectralID.lean`, `Paper1SpectralH.lean`	Lean 4 proof
Proposition S3.2, passive distinction with zero controlled quantity (machine-checked)	`research_extension/current_manuscript_lean_20261008/CurrentProofs/Paper1PassiveVariance.lean`	Lean 4 proof
Lemma S24.4, converse: pulse test errors and margin solvency (machine-checked)	`research_extension/current_manuscript_lean_20261008/CurrentProofs/Paper1CapitalConverse.lean`	Lean 4 proof
Propositions S5.1(ii) and S5.2(ii), Kalman-desk and tied-intensity lower bounds (machine-checked)	`research_extension/current_manuscript_lean_20261008/CurrentProofs/Paper1KalmanTied.lean`	Lean 4 proof
Supplement S18: post hoc robustness (futures prices, roll weeks, 26- and 52-week blocks)	`p11/robustness_cot.py`, `p11res/results/C4R_cot_robustness.json`	real data (post hoc)
Supplement S18 and Table S29: two-root test (2 of 8; smaller root -0.16 to 0.21)	`p11/two_root_test.py`, `p11res/results/C6M_cot_two_root.json` (specification `p11/prereg_memory/P11M_two_root.json`, e6633588…)	real data

Result	Source	Kind
Supplement S18 and Table S29: replication on 12 new markets (H1 12 of 12, H2 11 of 12, H3 3 of 12)	`p11/replication_cot.py`, result `C6R_cot_replication.json` (specification `p11/prereg_replication/P11R_cot_replication.json`, 591e71c9…)	real data
Figure 1 (paper in one figure)	`papers/paper1/figs/make_fig_summary.py` (inputs `p1_injection/results/I1_injection_test.json`, `p11res/results/C7_data_equivalence.json`)	registered results and post hoc calculation
Supplement S18, Table S26: price data needed to match the class reports (median factor 1,353; post hoc)	`p11/data_equivalence.py` → `p11res/results/C7_data_equivalence.json` (inputs `C4_cot_disclosure.json`, `C6R_extract.json`)	post hoc calculation
Supplement S19 and Table S29: October 2026 tests N2 (1 of 30) and N3 (5 of 30; 21 of 30; 30 of 30)	`p1_oct/oct_tests.py`, `p1_oct/results/zip/results/P1O_N2_feedback.json`, `P1O_N3_combined.json` (specification `p1_oct/prereg/P1O_october2026.json`, 72cbc8ba…; power check `P1O_power_check.json`)	real data
Supplement S18 and Table S29: class-flow test (3 of 8; $F \geq 10$ in 7; $w \neq 0$ in 6)	`p11/tagged_flow_test.py`, `p11res/results/C5F_cot_tagged_flow.json` (specification `p11/prereg_flow/P11F_cot_tagged_flow.json`, 052a63f2…)	real data
Finite-horizon decision appendix: primary decisions, cost–horizon sensitivities (819 rows), hard-pair requirements, assumption failures, cash-flow audit	`studies/finite_horizon_costs_20261029/run.py` → `results/final_full_20261030/`	

Result	Source	Kind
Finite-horizon appendix: independent checks and LP cross-check	`studies/finite_horizon_costs_20260129/validation/`	2601.22